ZHE DENG

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ACADEMIC EXPERIENCE

August 2023-Present	**SAINT JOSEPH'S UNIVERSITY**	**PHILADELPHIA, PENNSYLVANIA**
	Assistant Professor	
	Department of Decision and System Sciences, Haub School of Business	
	Co-Chair	
	AI Committee, Haub School of Business	
March 2022-March 2023	**CARNEGIE MELLON UNIVERSITY**	**PITTSBURGH, PENNSYLVANIA**
	Post-Doctoral Research Fellow	
	Heinz College of Information Systems and Public Policy	
August 2016-May 2022	**TEMPLE UNIVERSITY**	**PHILADELPHIA, PENNSYLVANIA**
	Research Assistant, Instructor	
	Fox School of Business	

EDUCATION

August 2016-May 2022	**TEMPLE UNIVERSITY**	**PHILADELPHIA, PENNSYLVANIA**
	Ph.D. in Management Information Systems	*(Presidential Fellowship)*
	Fox School of Business	
August 2012-May 2014	**UNIVERSITY OF FLORIDA**	**GAINESVILLE, FLORIDA**
	Master of Science in Information Systems & Operations Management	
	Hough Graduate School of Business, Warrington College of Business Administration	
August 2007	**NORTHWEST UNIVERSITY**	**XI'AN, CHINA**
	Bachelor of Science in Information Management & Information Systems	*(Magna Cum Laude)*
	School of Economics and Management	

RESEARCH INTERESTS

Topics: AI-enabled Information Systems; Human–AI Interaction and Decision Support Systems; Information Systems in Education and Healthcare; Emerging Technologies, Technology Policy, and Governance.

Methodologies Machine Learning and Deep Learning; Field Experiments; Econometric Analysis.

Publication

1. **Deng, Z.**, Z. Cheng, P. Ferreira, and P. A. Pavlou, From Smartphones to Smart Students: Learning versus Distraction with Mobile Devices in the Classroom and Student Success. Forthcoming. Information Systems Research.

WORKING PAPERS

1. Y. Chu, **Deng, Z.**, W. Wang, P. Ferreira. From Views to Learning: How Information Cues Transform Video-Based Online Education. Preparing for Management Science.
2. **Deng, Z.**, J. Jung, P. A. Pavlou, and Q. Sun, Would Unlimited Internet lead to Unlimited Learning? A Field Experiment of Internet Policy on Educational Outcomes. Preparing for journal submission.
3. **Deng, Z.**, S. Ayabakan, S. Kumar, and P. A. Pavlou. 2019, Patient-sharing Network, EHR Adoption, and Patient Outcomes: An Empirical Investigation. Preparing for journal submission.

WORKING IN PROGRESS

1. "Student Depression Detection: An Attention-based Bi-LSTM Approach Using Online and Offline Behavior Data" (With Guohou Shan, Konstantin Bauman, and Qian Sun)
2. "User-Generated Videos in Climate Change Education" (with Pedro Ferreira, Rodrigo Belo, and Yael Inbar)
3. "Student Reaction and Tutor Teaching Satisfaction: Evidence from Online Class Video Feeds" (with Pedro Ferreira)

PRESENTATIONS

1. **Deng, Z.**, P. Ferreira. 2024. Can Face Tell Learning? Evidence From Zoom Lectures. INFORMS Annual Meeting, Oct 20-23, 2024, Seattle, WA
2. Y. Chu, **Deng, Z.**, W. Wang, P. Ferreira. 2024. Multimodal Video Learning: Managing Cognitive Load Patterns using Guided Attention Graphs. INFORMS Annual Meeting, Oct 20-23, 2024, Seattle, WA
3. **Deng, Z**. 2024. Matching Supply with Demand: Data-Driven Methods in Supply Chain Analytics. PLMA Executive Education. Jun. 11, 2024, Philadelphia, PA
4. G. Shan, **Deng, Z.**, K. Bauman, and Q. Sun. 2022. Student Depression Detection: An Attention-based Bi-LSTM Approach Using Online and Offline Behavior Data. INFORMS Annual Meeting, Oct 16-19, 2022, Indianapolis, IN
5. **G. Shan**, Deng, Z.**,** K. Bauman, and Q. Sun. 2021. Student Depression Detection: An Attention-based Bi-LSTM Approach Using Online and Offline Behavior Data. China Summer Workshop on Information Management (CSWIM), July 4 – 6, 2021, Chongqing, China
6. G. Shan, **Deng, Z.**, K. Bauman, and Q. Sun. 2021. Student Depression Detection: An Attention-based Bi-LSTM Approach Using Online and Offline Behavior Data. Statistical Conference in E-Commerce Research (SCECR), June 17-18, 2021, Online
7. **Deng, Z.** 2021. Social Science Research Utilizing Online Traffic Auditing Data. Workshop on Online Traffic Data Science@Panabit Co. Ltd., March 31st, 2021, Beijing, China
8. G. Shan, **Deng, Z.**, K. Bauman, and Q. Sun. 2020. Student Depression Detection: An Attention-based Bi-LSTM Approach Using Online and Offline Behavior Data. Workshop on Information Technologies and Systems (WITS), December 16-18, 2020, Online
9. **Deng, Z.**, P. Ferreira, R. Belo, and Y. Inbar. 2020. Video Creation and Watching in Formal Learning Environments: A Randomized Field Experiment on Video-Based Peer Tutoring. Statistical Conference in E-Commerce Research (SCECR), June 18-20, 2020, Madrid, Spain
10. **Deng, Z.**, J. Jung, P. A. Pavlou and Q. Sun. 2020. Would Unlimited Internet lead to Unlimited Learning? A Field Experiment of Internet Policy on Educational Outcomes. Statistical Conference in E-Commerce Research (SCECR), June 18-20, 2020, Madrid, Spain
11. Deng, Z., Z. Cheng, **P. Ferreira**, and P. A. Pavlou. 2019. From Smartphones to Smart Students: Distraction versus Learning with Mobile Devices in the Classroom and Student Success. Guest Speaker Seminar at University of California, Berkeley, October 2, 2019, Berkeley, CA
12. **Deng, Z.** 2019. Data Mining and Knowledge Discovery. MBA Seminar Series at Northwest University, June 26, 2019, Xi'an, China
13. **Deng, Z.**, S. Ayabakan, S. Kumar, and P. A. Pavlou. 2019. Patient-sharing Network, EHR Adoption, and Healthcare Outcome: An Empirical Investigation. Statistical Conference in E-Commerce Research (SCECR), June 10-13, 2019, Hong Kong, China
14. Deng, Z., **S. Ayabakan**, S. Kumar, and P. A. Pavlou. 2019. Patient-sharing Network, EHR Adoption, and Patient Outcomes: An Empirical Investigation. Production and Operations Management Society (POMS) Annual Conference, May 2-6, 2019, Washington, D.C.
15. Deng, Z., Z. Cheng, **P. Ferreira**, and P. A. Pavlou. 2019. Mobile Device Policy, Attention Allocation, and Student Performance: Evidence from a Video-Tracking Randomized Experiment. Guest Speaker Seminar at University of Pittsburgh, April 19, 2019, Pittsburgh, PA
16. **Deng, Z.**, Z. Cheng, P. Ferreira, and P. A. Pavlou. 2018. Mobile Device Policy, Attention Allocation, and Student Performance: Evidence from a Video-Tracking Randomized Experiment. Workshop on Information Systems and Economics (WISE), December 17-18, 2018, San Francisco, CA
17. **Deng, Z.**, Z. Cheng, P. Ferreira, and P. A. Pavlou. 2018. Mobile Device Policy, Attention Allocation, and Student Performance:

Evidence from a Video-Tracking Randomized Experiment. Conference on Information Systems and Technology (CIST), November 3-4, 2018, Phoenix, Arizona

18. **Deng, Z.**, Z. Cheng, P. Ferreira, and P. A. Pavlou. 2018. Mobile Device Policy, Attention Allocation, and Student Performance: Evidence from a Video-Tracking Randomized Experiment. Conference on Digital Experimentation (CODE@MIT), October 27-28, 2018, Cambridge, MA
19. **Deng, Z.**, S. Ayabakan, S. Kumar, and P. A. Pavlou. 2018. Patient-sharing Network and Patient Outcomes: An Empirical Investigation. Conference on Health IT and Analytics (CHITA), October 19-20, 2018, Washington, D.C.
20. **Deng, Z.**, Z. Cheng, P. Ferreira, and P. A. Pavlou. 2018. Mobile Device Policy, Attention Allocation, and Student Performance: Evidence from a Video-Tracking Randomized Experiment. Statistical Conference in E-Commerce Research (SCECR), June 18-19, 2018, Rotterdam, Netherlands
21. **Deng, Z.**, S. Ayabakan, S. Kumar, and P. A. Pavlou. 2018. Physician Professional Network and Inpatient Outcomes: An Empirical Investigation. Production and Operations Management Society (POMS) Annual Conference, May 4-7, 2018, Houston, TX
22. **Deng, Z.**, Z. Cheng, P. Ferreira, and P. A. Pavlou. 2017. Mobile Distraction in Class: An Empirical Investigation on Smartphone Use Policies. Conference on Digital Experimentation (CODE@MIT), October 27-28, 2017, Cambridge, MA

TEACHING

1. **Artificial Intelligence**
    a. DSS 200 – AI in Business: Fall 2025, Spring 2026
2. **Business Intelligence and Analytics**
    a. DSS 315 – Business Intelligence and Analysis Concepts & Practices: Spring 2024, Spring 2025, Spring 2026
3. **Supply Chain Management**:
    a. DSS 335 - Foundations of Supply Chain Management: Fall 2023, Fall 2024, Spring 2025, Fall 2025
    b. DSS 465/720 - Supply Chain Analytics: Fall 2023, Spring 2024, Fall 2024, Fall 2025, Spring 2026
4. **Quality Control**:
    a. DSS 440/710 - Six Sigma Applications & Foundations: Spring 2024

HONORS AND AWARDS

1. **INFORMS Alpha Iota Delta and Omega Rho International Honor Societies, 2025**
   International honor societies of INFORMS recognizing outstanding academic achievement and research potential in operations research, management science, and information systems
2. **Presidential Fellowship, Temple University, 2016**
   University-wide, five-year fellowship awarded to top PhD students
3. **First Prize Scholarship (Magna Cum Laude), Northwest University, 2010**
   University-wide highest academic honor
4. **Outstanding Student Leader Award, Northwest University, 2009**
   Highest university-level leadership honor

FUNDING AND GRANTS

1. **Milton Lev Faculty Research Grants. Saint Joseph's University, $5,000**
   How Information Cues Transform Video-Based Online Education? Combining Randomized Experiments and Multimodal Machine Learning
2. **Summer Research Fund. Saint Joseph's University, $6,000**
   Would Unlimited Internet lead to Unlimited Learning? A Field Experiment of Internet Policy on Educational Outcomes.
3. **2020 21th Young Scholar Forum. Temple Univ. Grant, $2,500**

Student Depression Detection: An Attention-based Bi-LSTM Approach Using Online and Offline Behavior Data. (With G. Shan, K. Bauman, and Q. Sun)

4. **2020 Spring 2020 Seed Funding Forum. Temple Univ. Grant, $7,500**
   Would Unlimited Internet lead to Unlimited Learning? A Field Experiment of Internet Policy on Educational Outcomes. (With J. Jung, P. A. Pavlou, and Q. Sun)
5. **2020 20th Young Scholar Forum. Temple Univ. Grant, $1,000**
   Student Depression Detection Based on Online-offline Behaviors. (With G. Shan, K. Bauman and Q. Sun)
6. **2019 19th Young Scholar Forum. Temple Univ. Grant, $2,000**
   Unlimited Internet Use vs. Limited Performance? A Field Experiment on Education Outcomes. (with J. Jung, P. A. Pavlou, and Q. Sun)
7. **2018 17th Young Scholar Forum. Temple Univ. Grant, $2,000**
   Patient-sharing Network and Patient Outcomes: An Empirical Investigation. (With S. Ayabakan, S. Kumar, and P. A. Pavlou)
8. **2018 16th Young Scholar Forum. Temple Univ. Grant, $750**
   Physician Social Network and Patient Outcomes: An Empirical Investigation. (With S. Ayabakan, S. Kumar, and P. A. Pavlou)
9. **2017 14th Young Scholar Forum. Temple Univ. Grant, $2500**
   Mobile Distraction in Class: A Video-Tracking Field Experiment (With Z. Cheng, P. Ferreira, and P. A. Pavlou)
10. **2016 13th Young Scholar Forum. Temple Univ. Grant, $500**
    Smartphone: A Pandora's Box for Students in Class (With Z. Cheng, P. Ferreira, and P. A. Pavlou)

SERVICE

1. Session Chair – INFORMS Annual Meeting, 2024
2. Reviewer – MIS Quarterly 2019, 2020, 2021, 2022
3. Reviewer – Information Systems Research 2025
4. Reviewer – Electronic Commerce Research and Applications (ECRA) 2019, 2020
5. Reviewer – International Conference on Information Systems (ICIS) 2018, 2019, 2020, 2021, 2022, 2024, 2025
6. Reviewer – Conference on Information Systems and Technology (CIST) 2018, 2020, 2024
7. Reviewer – Workshop on Information Technologies and Systems (WITS) 2019
8. Reviewer – European Conference on Information Systems (ECIS) 2019
9. Reviewer – Pacific Asia Conference on Information Systems (ECIS) 2024
10. Reviewer – New England Association for Information Systems Conference 2022
11. External Researcher – Shaanxi Business College, China 2016, 2017, 2018, 2019, 2020, 2021, 2022
12. External Researcher – Kooledge.com 2018- 2022
13. External Researcher – Contemporary Education Technology Center, Northwest University, China 2019-2024
14. External Researcher – Nroad Net Co. Ltd. 2019, 2020, 2021, 2022
15. External Researcher – Panabit Co. Ltd. 2020-2024

SELECTED COURSEWORK

Micro-Economics, Game Theory, Information Economics

Statistics I, II, Econometrics I, II, III, IV

Management of Information Systems Seminars: Pro-seminar, E-commerce, Digital Innovation, Economics of IT

Operations Management Seminar: Continuous and Nonlinear Optimization Methods

Knowledge Discovery and Data Mining

SOFTWARE / PROGRAMMING SKILLS

Machine Learning and Data Science: Python (scikit-learn, PyTorch/TensorFlow), R

Data Analysis and Statistical Software: Python, R, Stata, SPSS, Wolfram Mathematica

Data Management and Visualization: SQL, Excel, Tableau
Programming and Web Technologies: C, C#, Java, ASP.NET, HTML